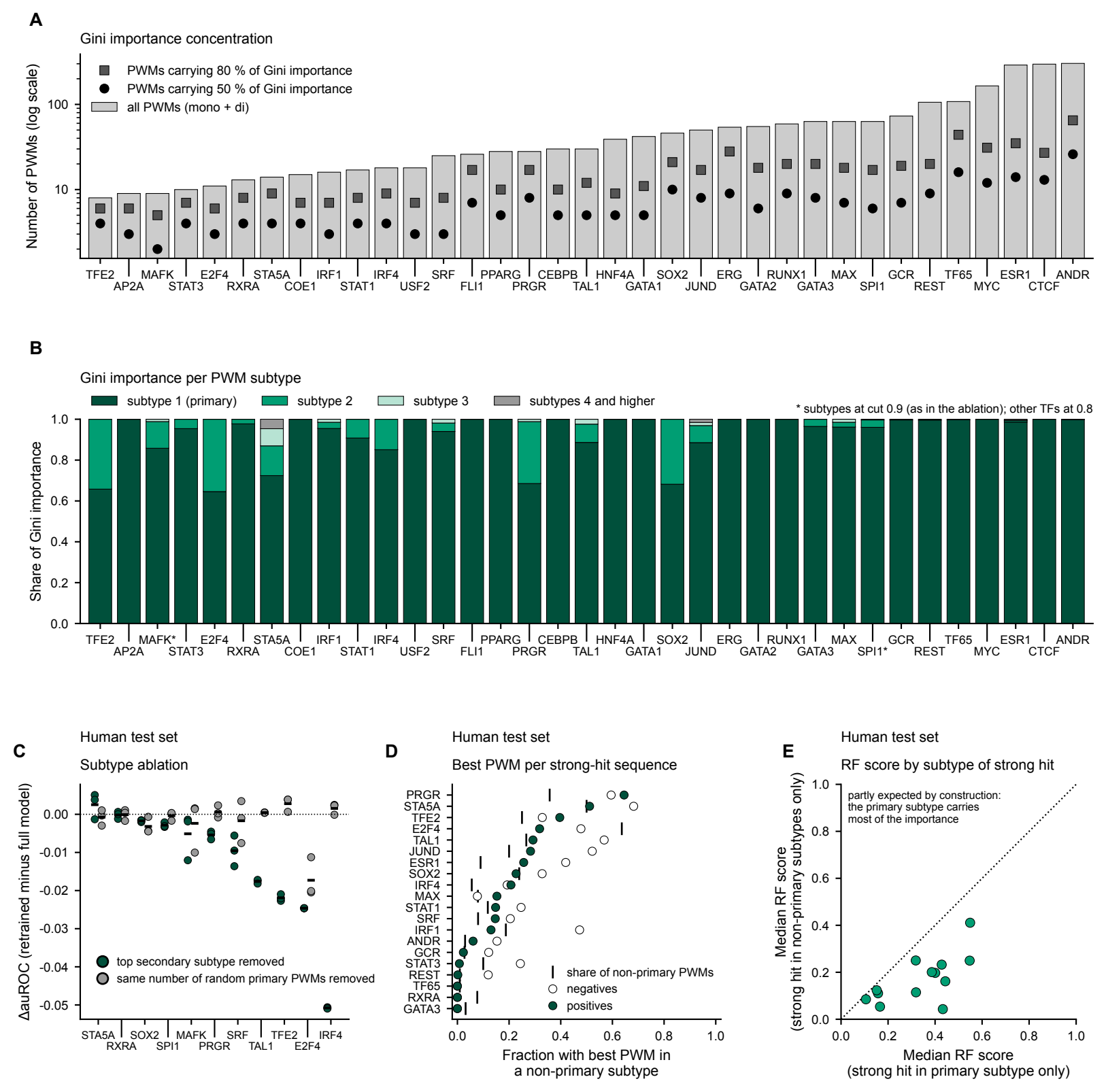

A
Gini importance concentration
PWMs carrying 80 % of Gini importance
PWMs carrying 50 % of Gini importance
all PWMs (mono + di)
Number of PWMs (log scale)
100
10
TFE2 AP2A MAFK STAT3 E2F4 RXRA STA5A COE1 IRF1 STAT1 IRF4 USF2 SRF FLI1 PPARG PRGR CEBPB TAL1 HNF4A GATA1 SOX2 JUND ERG GATA2 RUNX1 GATA3 MAX SPI1 GCR REST TF65 MYC ESR1 CTCF ANDR
B
Gini importance per PWM subtype
subtype 1 (primary)
subtype 2
subtype 3
subtypes 4 and higher
* subtypes at cut 0.9 (as in the ablation); other TFs at 0.8
Share of Gini importance
1.0
0.8
0.6
0.4
0.2
0.0
TFE2 AP2A MAFK* STAT3 E2F4 RXRA STA5A COE1 IRF1 STAT1 IRF4 USF2 SRF FLI1 PPARG PRGR CEBPB TAL1 HNF4A GATA1 SOX2 JUND ERG GATA2 RUNX1 GATA3 MAX SPI1* GCR REST TF65 MYC ESR1 CTCF ANDR
C
Human test set
Subtype ablation
ΔauROC (retrained minus full model)
0.00
-0.01
-0.02
-0.03
-0.04
-0.05
top secondary subtype removed
same number of random primary PWMs removed
STA5A RXRA SOX2 SPI1 MAFK PRGR SRF TAL1 TFE2 E2F4 IRF4
D
Human test set
Best PWM per strong-hit sequence
PRGR STA5A TFE2 E2F4 TAL1 JUND ESR1 SOX2 IRF4 MAX STAT1 SRF IRF1 ANDR GCR STAT3 REST TF65 RXRA GATA3
share of non-primary PWMs
negatives
positives
0.0 0.2 0.4 0.6 0.8 1.0
Fraction with best PWM in a non-primary subtype
E
Human test set
RF score by subtype of strong hit
partly expected by construction: the primary subtype carries most of the importance
Median RF score (strong hit in non-primary subtypes only)
1.0
0.8
0.6
0.4
0.2
0.0
0.0 0.2 0.4 0.6 0.8 1.0
Median RF score (strong hit in primary subtype only)